Paper VIII
Sixth Lean Verification Block

Small-Core Optimal Recompression
and Completion of the Finite Register

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Scope and result

Theorem 5.6, the last pending manuscript-proved item, is now covered. The finite register contains **56 of 56 checked items**. Four classical inputs remain separately tracked. This block adds **79 named theorem declarations in ten modules**, giving **392 declarations in 40 modules**. The total contains 373 Paper VIII declarations and 19 in the unchanged reused quantum-packet foundation. Definitions and generated declarations are not added to that theorem count.

The proof constructs compact orthonormal factors from the actual input column spaces. Given an *ordered thin singular-value decomposition of the actual small core*, it constructs thin square-root factors, proves their rank bound and global optimality in both the spectral and Frobenius norms, and proves the exact error formulas. The diagonal best-approximation inequalities are proved in this release, not assumed through an optimality certificate.

Fresh cumulative check	Result
Active Lean modules compiled	40/40
Named theorem dependency audits	392/392
Separately counted construction audits	66/66
Deliberately invalid Lean proof controls	Both rejected
Verifier, compatibility and sixth-block unit tests	43 passed
Independent numerical recompression cases	638 passed

Factorization boundary. As in the manuscript's instruction to take an SVD, its exact reconstruction, orthonormality, ordering and nonnegativity equations are supplied. No optimality or tail-error conclusion is supplied. General SVD existence, a floating-point SVD algorithm, and identification with the separately named Mathlib `singularValues` API are not independently formalized here. The exact column-space construction is not a numerical QR implementation.

All 30 upstream proof modules, the original manuscript assets and five earlier supplements remain byte-identical. The source archive is based on the 313-declaration fifth block, not the earlier same-named 298-declaration branch.

1. Compact factors from actual column spaces

Modules CompactRangeFactors and SmallCoreTransfer

Let

$$L \in \mathbb{C}^{m\times t}, \qquad R \in \mathbb{C}^{n\times t}, \qquad A = LR^*.$$

Use the ordinary Euclidean inner products on the coordinate spaces. Define

$$S_L = \operatorname{range}(L), \quad r_L = \dim S_L, \qquad S_R = \operatorname{range}(R), \quad r_R = \dim S_R.$$

The implementation constructs an orthonormal basis of each actual range subspace. Let Q_L and Q_R be the matrices of the corresponding isometric inclusions. Express the range-restricted input maps in these bases to obtain R_L, R_R.

The checked identities are

$$\boxed{Q_L^*Q_L = I, \quad L = Q_L R_L, \quad R_L = Q_L^* L,}$$

with the analogous identities on the right. The dimension theorem is not an input:

$$\boxed{r_L = \operatorname{rank} L, \qquad r_R = \operatorname{rank} R.}$$

The coordinate factors have full row rank. The construction therefore handles rank-deficient input factors without retaining artificial column directions.

Define the small core by actual multiplication,

$$G = R_L R_R^* \in \mathbb{C}^{r_L\times r_R}.$$

Finite matrix associativity and adjoint identities give

$$\boxed{A = Q_L G Q_R^*, \qquad \operatorname{rank} G = \operatorname{rank} A.}$$

The rank identity follows from both multiplication inequalities and the left inverse obtained by compression. It is not inferred from agreeing dimensions.

Zero factors and empty spaces. No positive-rank or nonempty-coordinate hypothesis is imposed. If a range is zero, its orthonormal basis has an empty index type. The factorization uses the resulting rectangular empty matrices. A separate theorem proves that the compact dimension is zero exactly when the original factor is zero.

Relation to QR. The argument only needs orthonormal columns and exact reconstruction. A compact QR factorization supplies these properties, but triangularity of its coordinate factor is irrelevant to the recompression theorem. The formal construction uses an exact orthonormal range basis; it does not claim Householder steps, pivot selection or floating-point stability. A supplied noncompact orthonormal factorization is also covered by the general transfer theorem, including extra zero singular directions.

2. Why a small-core optimum is a global optimum

Module SmallCoreNorms; arbitrary ambient competitors

For rectangular isometries Q, V define

$$\mathcal{L}(G) = QGV^*, \qquad \mathcal{C}(B) = Q^*BV, \qquad Q^*Q = I, \quad V^*V = I.$$

The following statements are proved for the actual matrix rank and actual matrix norms:

$$\mathcal{C}(\mathcal{L}(G)) = G, \qquad \operatorname{rank}\mathcal{L}(G) = \operatorname{rank} G, \qquad \operatorname{rank}\mathcal{C}(B) \le \operatorname{rank} B,$$

$$\boxed{\|\mathcal{L}(G)\|_\nu = \|G\|_\nu, \qquad \|\mathcal{C}(B)\|_\nu \le \|B\|_\nu, \quad \nu \in \{2, \mathrm{F}\}.}$$

For the spectral norm, the proof uses the Gram matrix identity

$$(QG)^*(QG) = G^*G$$

and the C^*-norm square identity, together with right-side adjoint symmetry. Compression follows from submultiplicativity and the isometry norm bound, including empty spaces.

For the Frobenius norm, trace cyclicity proves left and right isometric invariance. The proof also establishes the projection identity

$$\|B - QQ^*B\|_{\mathrm{F}}^2 = \|B\|_{\mathrm{F}}^2 - \|Q^*B\|_{\mathrm{F}}^2.$$

Nonnegativity of the left side gives left compression contractivity; the right version follows by adjoints. No norm-preservation or contractivity hypothesis is inserted.

Suppose G_k is a best rank-at-most-k approximation to G in either norm. Take an *arbitrary* ambient matrix B with $\operatorname{rank} B \le k$. Compression is an admissible small-core competitor and

$$\begin{aligned} \|\mathcal{L}(G) - \mathcal{L}(G_k)\|_\nu &= \|G - G_k\|_\nu \\ &\le \|G - \mathcal{C}(B)\|_\nu \\ &= \|\mathcal{C}(\mathcal{L}(G) - B)\|_\nu \\ &\le \|\mathcal{L}(G) - B\|_\nu. \end{aligned}$$

This is the key global bridge: B is not required to lie in the supplied row and column spaces. The next two pages discharge the intermediate small-core optimality obligation, rather than leaving it as a hypothesis in the final theorem.

3. Spectral optimality from a kernel witness

Modules DiagonalTruncation and SpectralOptimality

Let $\sigma_1 \geq \cdots \geq \sigma_d \geq 0$ and

$$D = \operatorname{diag}(\sigma_1, \ldots, \sigma_d), \qquad D_k = \operatorname{diag}(\sigma_1, \ldots, \sigma_{\min(k,d)}, 0, \ldots, 0).$$

Lean uses indices beginning at zero; the formulas on this page use the conventional one-based notation. The rank of D_k is proved to be at most k, even when retained singular values are zero.

For $k < d$, take any $d \times d$ matrix B of rank at most k. Restrict its domain to the first $k+1$ coordinates using the actual coordinate inclusion J. Since

$$\operatorname{rank}(BJ) \leq \operatorname{rank} B \leq k < k+1,$$

rank-nullity supplies a nonzero vector z with $BJz = 0$. The vector $y = Jz$ is nonzero, satisfies $By = 0$ and is supported in the first $k+1$ coordinates. The injection and the support property are checked separately.

Monotonicity of the diagonal entries gives

$$\|Dy\|^2 = \sum_{i=1}^{k+1} \sigma_i^2 |y_i|^2 \geq \sigma_{k+1}^2 \|y\|^2 .$$

Because $(D - B)y = Dy$, the operator-norm inequality implies

$$\sigma_{k+1} \|y\| \leq \|(D - B)y\| \leq \|D - B\|_2 \|y\| .$$

Cancellation of the strictly positive norm of y proves

$$\boxed{\sigma_{k+1} \leq \|D - B\|_2 .}$$

The norm of the actual diagonal residual $D - D_k$ equals σ_{k+1}, so D_k attains the lower bound. If $k \geq d$, the residual is zero and optimality follows from norm nonnegativity.

No optimality oracle. The nonzero kernel vector, its coordinate support and the diagonal norm bound are all derived. No Eckart–Young inequality, best-rank property or singular-tail formula is assumed. The final rectangular SVD result follows by the isometric transfer proved on the preceding page.

4. Frobenius optimality from kernel weights

Modules KernelFrame and FrobeniusOptimality

For an arbitrary competitor B with $\operatorname{rank} B \le k$, construct an orthonormal matrix Q spanning its actual kernel. Then

$$BQ = 0, \qquad Q^*Q = I, \qquad q = \dim\ker B = d - \operatorname{rank} B.$$

Define the row weights

$$w_i = \sum_{j=1}^{q} |Q_{ij}|^2.$$

The proof derives

$$\boxed{0 \le w_i \le 1, \qquad \sum_i w_i = q \ge \max(d-k, 0).}$$

The upper bound uses contraction of Q^* on a coordinate vector. The weight sum follows from the trace of Q^*Q and rank-nullity.

For any nonnegative decreasing sequence a_i, weights with these properties satisfy

$$\sum_i a_i w_i \ge \sum_{i>k} a_i.$$

This finite weighted inequality is itself proved. For $k < d$, use $\mu = a_{k+1}$ and compare the summands to tail weights $v_i = 0$ for $i \le k$ and $v_i = 1$ otherwise:

$$a_i v_i + \mu(w_i - v_i) \le a_i w_i.$$

For a retained coordinate the inequality uses $a_i \ge \mu$ and $w_i \ge 0$. For a tail coordinate it uses $a_i \le \mu$ and $w_i \le 1$. Summation and $\sum w_i \ge \sum v_i$ give the result.

Set $a_i = \sigma_i^2$. Right compression cannot increase the Frobenius norm, and $BQ = 0$, hence

$$\begin{aligned} \|D - B\|_{\mathrm{F}}^2 &\ge \|(D-B)Q\|_{\mathrm{F}}^2 \\ &= \|DQ\|_{\mathrm{F}}^2 = \sum_i \sigma_i^2 w_i \ge \sum_{i>k} \sigma_i^2. \end{aligned}$$

Direct calculation of $D - D_k$ attains equality. Thus the best-rank Frobenius assertion is proved from kernel geometry and finite sums, independently of the spectral lower-bound proof. Repeated and zero singular values require no uniqueness claim.

5. Thin output factors and the final theorem

Modules CompactTruncation and SmallCoreRecompression

For the actual core $G = R_L R_R^*$, supply ordered thin SVD data

$$G = W \operatorname{diag}(\sigma) Z^*, \qquad W^*W = I, \qquad Z^*Z = I, \qquad \sigma_1 \geq \cdots \geq \sigma_d \geq 0.$$

These are precisely the fields of the formal SVD structure. It contains no rank-optimality or tail-error field. A derived theorem checks the Gram eigenvector equation

$$G^*GZ = Z \operatorname{diag}(\sigma_1^2, \ldots, \sigma_d^2).$$

Let $h = \min(d, k)$ and retain exactly the first h column indices. The rectangular square-root mask has entries

$$M_{ij} = \begin{cases} \sqrt{\sigma_j}, & i = j, \\ 0, & i \neq j, \end{cases} \qquad 1 \leq i \leq d, \quad 1 \leq j \leq h.$$

Lean proves its index cardinality and $MM^* = D_k$. Define

$$\boxed{L_k = Q_L W M, \qquad R_k = Q_R Z M.}$$

The product equals $Q_L W D_k Z^* Q_R^*$ and has rank at most k. The principal theorem proves, for every ambient B with $\operatorname{rank} B \leq k$,

$$\boxed{\|LR^* - L_k R_k^*\|_\nu \leq \|LR^* - B\|_\nu, \quad \nu \in \{2, \mathrm{F}\}.}$$

It also gives the exact errors

$$\|LR^* - L_k R_k^*\|_2 = \begin{cases} \sigma_{k+1}, & k < d, \\ 0, & k \geq d, \end{cases} \qquad \|LR^* - L_k R_k^*\|_{\mathrm{F}} = \left(\sum_{i>k} \sigma_i^2 \right)^{1/2}.$$

Both the squared and square-root Frobenius formulas are checked. All these statements are combined in

`KomorebiPaperVIII.SmallCore.small_core_optimal_recompression`.

Endpoints. Rank zero produces the zero output; $k \geq d$ reconstructs the original product. An empty SVD certificate is constructed for a zero core, and diagonal SVD certificates are constructed explicitly. Zero singular values are never inverted. At repeated singular values the result asserts optimality, not uniqueness of the minimizer.

6. Source map and verification

One completed manuscript item; 79 supporting declarations

Module	Main purpose	Theorems
SmallCoreNorms	Isometric norms, compression, rank	19
CompactRangeFactors	Actual compact column factors	7
SmallCoreTransfer	Global best-rank transfer	7
KernelFrame	Actual kernel and row weights	8
DiagonalTruncation	Rank and exact diagonal errors	9
FrobeniusOptimality	Weighted-tail lower bound	5
SpectralOptimality	Prefix-kernel lower bound	7
CompactTruncation	Thin square-root factors	5
SmallCoreRecompression	Main theorem and Gram relation	6
RecompressionEndpoints	Zero and full-retention cases	6
Total	Theorem 5.6 and its support	79

Cumulative Lean run. All 40 local modules were compiled in a newly created output directory under Lean 4.34.0-rc2. Each exited with code zero and warnings were fatal. The 392 named theorem reports and 66 separately counted construction reports match the inventory exactly. Dependencies are confined to `propext`, `Classical.choice` and `Quot.sound`, or subsets thereof. False equality and admitted-proof controls were both rejected.

Pinned dependency. The actual Mathlib Git revision and manifest revision both match

`aeb72f1e9981b6d35b426b16dca7759be3e7ba06`. The supplied compiled Mathlib cache was reused. It was not independently rebuilt or rechecked. Recovery verified all seven offline archive parts and the combined archive hash.

Infrastructure. Forty-three unit tests passed: 17 verifier tests, 12 fifth-block compatibility tests updated to the current totals, and 14 sixth-block tests. The original fifth-block fixed-count test file and metadata are preserved. Exact audit coverage, dependency order, no-bypass scans, classical-input boundaries and absence of optimality fields in the SVD structure are tested.

Failures retained. Of 31 captured development checks, 21 failed before correction and ten accepted drafts supplied the new modules. Failures concerned proof elaboration, finite-index/cardinality APIs, dependent rewrites, norm scopes and warning diagnostics. Their inactive source snapshots and logs are retained separately from the successful fresh cumulative run. An initial interactive runtime launch was unsupported; noninteractive assembly succeeded.

7. Regressions, reproduction and claim boundary

Exact mathematical guarantees versus numerical implementations

Independent numerical checks. A deterministic NumPy complex128 mirror tested 182 input-factor pairs at 638 truncation cases. The cases include rectangular matrices, rank-deficient and zero factors, empty spaces, repeated singular values, rank zero and ranks beyond the core width. Comparison used both the small core and an independently computed ambient SVD reference. There were 3,190 arbitrary rank-bounded competitors, nine invalid-input rejection controls and two checks demonstrating failure when essential assumptions are removed.

The largest relative tail/reconstruction discrepancy was below 3.7×10^{-15}, against a declared 2×10^{-9} tolerance. The mirror uses numerical rank thresholds and a numerical SVD to construct column bases. It is not compiled or extracted Lean code, a proof for all inputs, or a GPU benchmark. The formal result uses exact ranks and exact factorization equations.

Reproduce from the extracted source root.

```
python3 scripts/verify_integrity.py
python3 -m unittest discover -s scripts -p 'test_*.py' -v
python3 scripts/verify.py --kit /path/to/portable-kit \
    --out verification/recheck --timeout 240
python3 scripts/run_sixth_regressions.py \
    --out verification/recheck_numerical.json
```

The unit tests and numerical mirror require NumPy. The Lean verification path needs Python 3.10 or later and the separately supplied pinned Linux x86-64 kit. Local compiled interfaces are regenerated. The runner also supports bounded batches with `-max-checks` and hash-checked `-resume`. A partial checkpoint is not a pass.

What is complete. The 56-item manuscript-proved register is covered under the recorded hypotheses. The original theorem says to take orthonormal factors and a core SVD. The new construction and proof discharge the recompression conclusion from those exact factorization equations. In particular, no best-approximation assertion or error formula is hidden in an input certificate.

What is not claimed. General SVD existence or a numerical SVD implementation is not newly formalized. The four classical entries remain separately tracked; the ordered-SVD-data optimality argument now has checked support, while the CG algorithm theorem, floating-point product model and simultaneous-decomposition input are not newly proved in full. No new generic matrix-multiplication exponent, floating-point stability theorem or hardware speedup follows from this release.

The original manuscript, bibliography, source package and all five earlier supplements are preserved. Historical CPU/GPU measurements, earlier compiler execution suites and Papers I–VII in full were not rerun. Prose-to-formal correspondence is an explicit mathematical review, not an automated theorem equating a sentence with a Lean type.